Exercise 9.1

(1) $\simeq 0.\overline{3}$ (2) 0.25 (3) $\frac{1}{2\sqrt{3}} \simeq 0.288$ (4) $\simeq 0.25$

(5) 1 (6) 0 (7) 1 (8) 3

(9) 2 (10) 3 (11) does not exist (12) does not exist

(13) 0 (14) 1 (15) does not exist

(16) except at $x_0 = 4$ (17) except at $x_0 = \pi$ (19) $f(8^-) = f(8^+) = 25$

(20) No (21) $f(2)$ cannot be concluded (22) 6, 6 (23) does not exist

Exercise 9.2

(1) 32 (2) $\frac{m}{n}$ (3) 108 (4) $\frac{1}{2\sqrt{x}}$

(5) $\frac{1}{6}$ (6) $-\frac{1}{4}$ (7) 3 (8) 4

(9) $\frac{1}{2}$ (10) $-\frac{1}{4}$ (11) $-\frac{3}{4}\sqrt[3]{4}$ (12) 0

(13) $f(x) \to -\infty$ as $x \to 0$ (limit does not exist) (14) $\frac{1}{4}$ (15) $\frac{1}{4a\sqrt{a-b}}$

Exercise 9.3

(1) (i) $f(-2) \to \infty$ as $x \to -2^-$, $f(-2) \to -\infty$ as $x \to -2^+$

(ii) $f\left(\frac{\pi}{2}\right) \to \infty$ as $x \to \frac{\pi}{2}^-$, $f\left(\frac{\pi}{2}\right) \to -\infty$ as $x \to \frac{\pi}{2}^+$

(2) $f(3) \to -\infty$ as $x \to 3^-$, $f(3) \to \infty$ as $x \to 3^+$ (3) $f(x) \to \infty$ as $x \to \infty$

(4) 0 (5) $f(x) \to \infty$ as $x \to \infty$ (6) -1 (7) $\frac{1}{4}$

(9) $\frac{1}{\alpha}$ (10) 30

Exercise 9.4

(1) e^7 (2) $e^{\frac{1}{3}}$ (3) 1 (4) $\frac{1}{e^8}$ (5) e^3

(6) $\frac{1}{8}$ (7) $\frac{\alpha}{\beta}$ (8) $\frac{2}{5}$ (9) $\begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m > n \\ f(\alpha) \to \infty \text{ as } \alpha \to 0 & \text{if } m < n \end{cases}$

(10) $2\cos a$ (11) $\frac{b}{a}$ (12) $\frac{2}{3}$ (13) $\frac{1}{2}$ (14) 2

(15) $\log\frac{2}{3}$ (16) $\log 9$ (17) $\frac{1}{2}$ (18) $\log 3 - 1$ (19) a

(20) $-\frac{3}{2}$ (21) e^2 (22) $\frac{1}{4\sqrt{2}}$ (23) 1 (24) e^2

(25) 2 (26) $a - b$ (27) $\frac{1}{2}$ (28) $\frac{1}{2}$

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